

TASK-3

Code:

```
#define SOIL_PIN A0
#define WATER_PIN A1
#define TEMP_PIN A2

#define RELAY_PIN 6
#define LED_PIN 9 // optional indicator LED

void setup() {
    Serial.begin(9600);

    pinMode(RELAY_PIN, OUTPUT);
    pinMode(LED_PIN, OUTPUT);

    digitalWrite(RELAY_PIN, LOW);
    digitalWrite(LED_PIN, LOW);
}

void loop() {

    int soilValue = analogRead(SOIL_PIN); int
    waterLevel = analogRead(WATER_PIN); int
    tempValue = analogRead(TEMP_PIN); float
    voltage = tempValue * (5.0 / 1023.0); float
    temperature = (voltage - 0.5) * 100.0;
```

```
Serial.println("-----");

Serial.print("Soil Moisture : ");
Serial.println(soilValue);

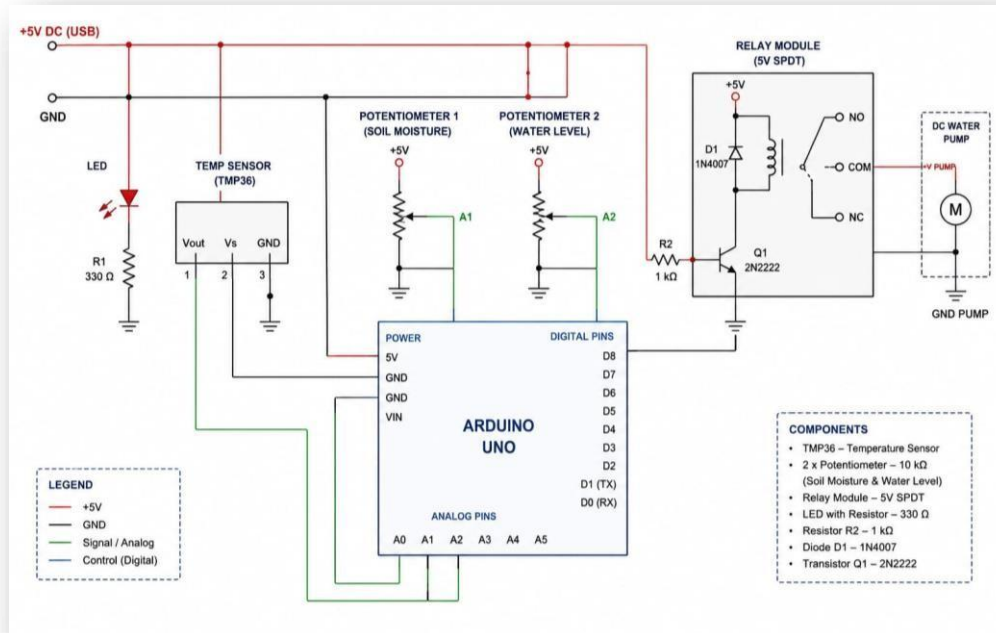
Serial.print("Water Level  : ");
Serial.println(waterLevel);

Serial.print("Temperature  : ");
Serial.print(temperature);
Serial.println(" °C");

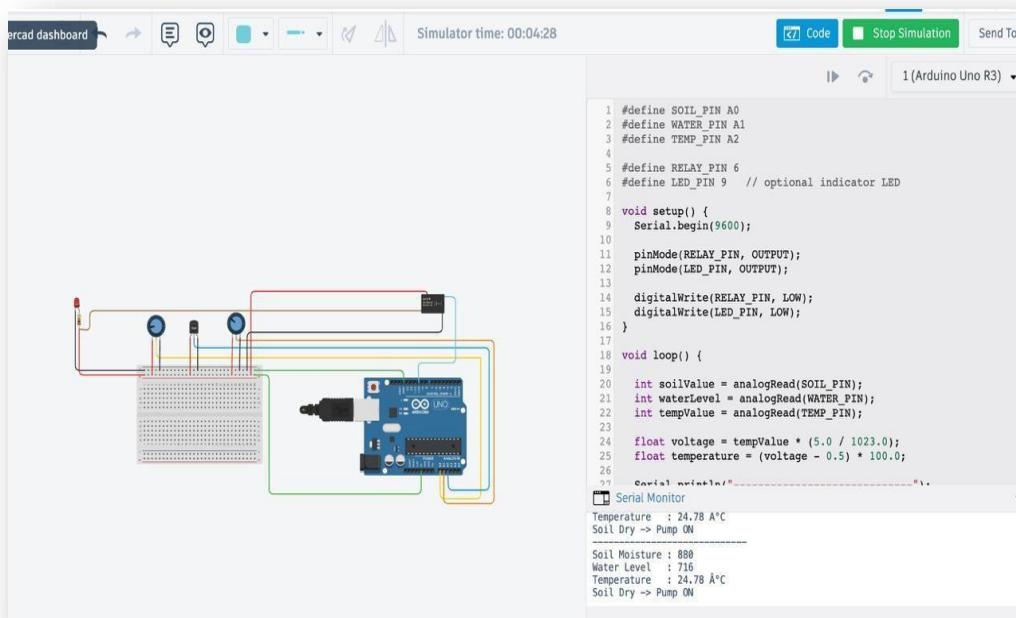
// SPDT Relay control (same logic as normal relay)  if (soilValue > 600) {
digitalWrite(RELAY_PIN, HIGH); // activates coil → switches COM to NO
digitalWrite(LED_PIN, HIGH);
  Serial.println("Soil Dry -> Pump ON");
} else {
  digitalWrite(RELAY_PIN, LOW); // COM switches back to NC
digitalWrite(LED_PIN, LOW);
  Serial.println("Soil Wet -> Pump OFF");
}

delay(2000);
}
```

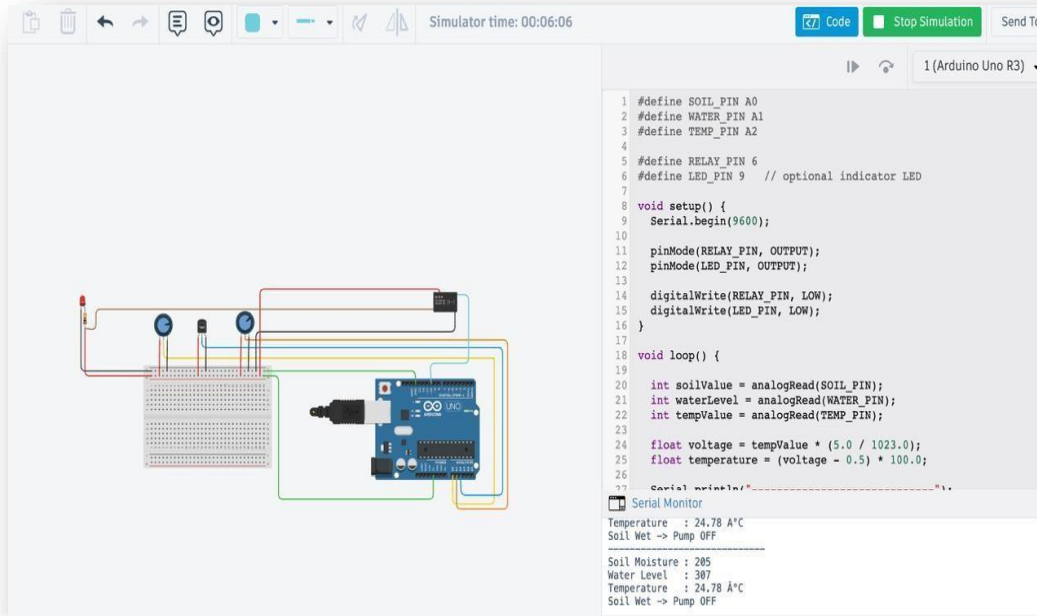
Circuit: IoT SOIL MONITORING SYSTEM



Outputs: POWER ON



POWER OFF



The screenshot displays the Arduino IDE interface during a simulation. On the left, a circuit diagram shows an Arduino Uno R3 connected to a breadboard. The breadboard contains a 555 timer, two LEDs, and several resistors. Wires connect the Arduino's digital and power pins to the breadboard components. On the right, the code editor shows the following C++ code:

```
1 #define SOIL_PIN A0
2 #define WATER_PIN A1
3 #define TEMP_PIN A2
4
5 #define RELAY_PIN 6
6 #define LED_PIN 9 // optional indicator LED
7
8 void setup() {
9   Serial.begin(9600);
10
11   pinMode(RELAY_PIN, OUTPUT);
12   pinMode(LED_PIN, OUTPUT);
13
14   digitalWrite(RELAY_PIN, LOW);
15   digitalWrite(LED_PIN, LOW);
16 }
17
18 void loop() {
19
20   int soilValue = analogRead(SOIL_PIN);
21   int waterLevel = analogRead(WATER_PIN);
22   int tempValue = analogRead(TEMP_PIN);
23
24   float voltage = tempValue * (5.0 / 1023.0);
25   float temperature = (voltage - 0.5) * 100.0;
26
27   Serial.println("-----");
28 }
```

Below the code editor, the Serial Monitor window is open, displaying the following output:

```
Temperature : 24.78 A°C
Soil Wet -> Pump OFF
-----
Soil Moisture : 205
Water Level : 307
Temperature : 24.78 A°C
Soil Wet -> Pump OFF
```